

- (c) (i) • Stand the bottle upright on the bench.
- The height of the water in the bottle is  $h$ , as shown in Fig. 2.5.

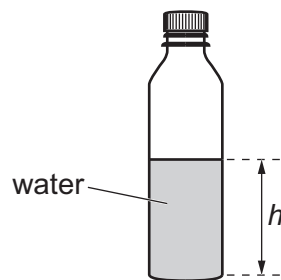


Fig. 2.5

Measure and record  $h$ .

$h =$  .....

- The time for the bottle to roll distance  $L$  on the rules is  $t$ .

Take measurements to determine  $t$ .

$t =$  ..... [3]

- (ii) A value for the acceleration  $a$  of the bottle is given by

$$a = \frac{2L}{t^2}.$$

Calculate  $a$ .

$a =$  ..... [1]

- (iii) Justify the number of significant figures that you have given for your value of  $a$ .

.....

.....

..... [1]

